

APPLICATIONS OF LINEAR ALGEBRA: PROBLEM SET 4

RENTO SAIJO

May 12, 2026

PROBLEM 1

A student runs an experiment six times in an attempt to obtain an equation relating two physical quantities x and y . For $x = 1, 2, 4, 6, 8, 10$ units, the experiments result in corresponding y values of 3, 3, 4, 6, 7, 8.

- (a) Find and compare the graphs of the following:
- (i) the least squares line;
 - (ii) the least squares quadratic polynomial; and
 - (iii) the interpolating polynomial.
- (b) Which do you think is the most likely theoretical model for this data? Why?

- (a) We start with the least squares line $y = mx + b$. If

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 4 & 1 \\ 6 & 1 \\ 8 & 1 \\ 10 & 1 \end{pmatrix}, \quad \vec{y} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 6 \\ 7 \\ 8 \end{pmatrix},$$

then the normal equations are

$$A^T A \begin{pmatrix} m \\ b \end{pmatrix} = A^T \vec{y}.$$

Here,

$$A^T A = \begin{pmatrix} 221 & 31 \\ 31 & 6 \end{pmatrix}, \quad A^T \vec{y} = \begin{pmatrix} 197 \\ 31 \end{pmatrix}.$$

So we solve

$$\begin{pmatrix} 221 & 31 \\ 31 & 6 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 197 \\ 31 \end{pmatrix},$$

which gives

$$m = \frac{221}{365}, \quad b = \frac{744}{365}.$$

Therefore, the least squares line is

$$y = \frac{221}{365}x + \frac{744}{365}.$$

Now, do the same thing for a quadratic model

$$y = ax^2 + bx + c.$$

This time,

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 16 & 4 & 1 \\ 36 & 6 & 1 \\ 64 & 8 & 1 \\ 100 & 10 & 1 \end{pmatrix}, \quad \vec{y} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 6 \\ 7 \\ 8 \end{pmatrix},$$

so the normal equations become

$$A^T A \begin{pmatrix} a \\ b \\ c \end{pmatrix} = A^T \vec{y}.$$

We compute

$$A^T A = \begin{pmatrix} 15665 & 1801 & 221 \\ 1801 & 221 & 31 \\ 221 & 31 & 6 \end{pmatrix}, \quad A^T \vec{y} = \begin{pmatrix} 1543 \\ 197 \\ 31 \end{pmatrix}.$$

Solving this system gives

$$a = \frac{25}{4652}, \quad b = \frac{12729}{23260}, \quad c = \frac{24903}{11630}.$$

So the least squares quadratic is

$$y = \frac{25}{4652}x^2 + \frac{12729}{23260}x + \frac{24903}{11630}.$$

For the interpolating polynomial, the x -values are all distinct, so the Vandermonde matrix is regular. Therefore, there is a unique interpolating polynomial of degree at most 5. Write

$$p(x) = c_5x^5 + c_4x^4 + c_3x^3 + c_2x^2 + c_1x + c_0.$$

Then, the coefficients satisfy

$$\begin{pmatrix} 1^5 & 1^4 & 1^3 & 1^2 & 1 & 1 \\ 2^5 & 2^4 & 2^3 & 2^2 & 2 & 1 \\ 4^5 & 4^4 & 4^3 & 4^2 & 4 & 1 \\ 6^5 & 6^4 & 6^3 & 6^2 & 6 & 1 \\ 8^5 & 8^4 & 8^3 & 8^2 & 8 & 1 \\ 10^5 & 10^4 & 10^3 & 10^2 & 10 & 1 \end{pmatrix} \begin{pmatrix} c_5 \\ c_4 \\ c_3 \\ c_2 \\ c_1 \\ c_0 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 6 \\ 7 \\ 8 \end{pmatrix}.$$

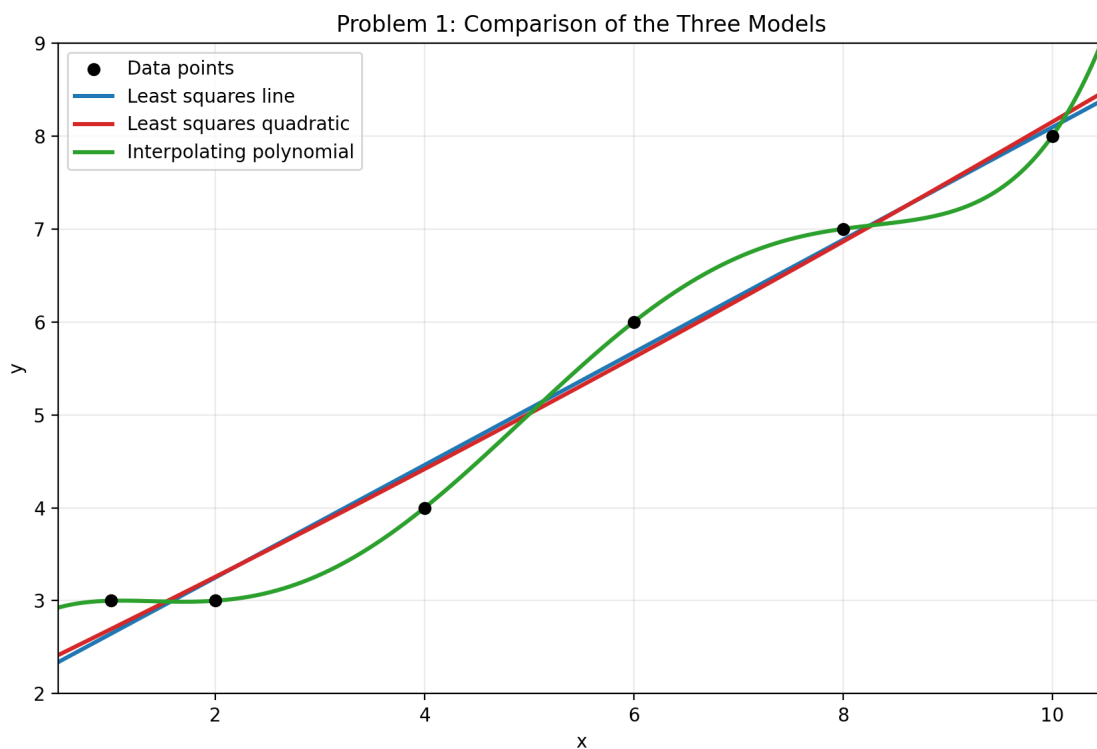
Solving this system gives

$$p(x) = \frac{169}{120960}x^5 - \frac{275}{8064}x^4 + \frac{419}{1512}x^3 - \frac{535}{672}x^2 + \frac{6931}{7560}x + \frac{166}{63}.$$

On the graph below (via Python), the least squares line and the least squares quadratic are very close to each other. The quadratic does bend upward a little bit, but the bend is small because the coefficient of x^2 is only

$$\frac{25}{4652} \approx 0.005374.$$

The interpolating polynomial is the one that passes through every single data point exactly, but it also has the most freedom to wiggle between the data points.



- (b) The least squares line seems like the most likely theoretical model. The main reason is that the data has a pretty clear overall linear upward trend, and the quadratic fit only improves the least squares error by a very small amount. For the line, the fitted values at the six data points are

$$\frac{193}{73}, \quad \frac{1186}{365}, \quad \frac{1628}{365}, \quad \frac{414}{73}, \quad \frac{2512}{365}, \quad \frac{2954}{365}.$$

So the residuals are

$$\begin{aligned} 3 - \frac{193}{73} &= \frac{26}{73}, & 3 - \frac{1186}{365} &= -\frac{91}{365}, & 4 - \frac{1628}{365} &= -\frac{168}{365}, \\ 6 - \frac{414}{73} &= \frac{24}{73}, & 7 - \frac{2512}{365} &= \frac{43}{365}, & 8 - \frac{2954}{365} &= -\frac{34}{365}. \end{aligned}$$

Therefore,

$$\text{SSE for the line} = \left(\frac{26}{73}\right)^2 + \left(-\frac{91}{365}\right)^2 + \left(-\frac{168}{365}\right)^2 + \left(\frac{24}{73}\right)^2 + \left(\frac{43}{365}\right)^2 + \left(-\frac{34}{365}\right)^2 = \frac{194}{365} \approx 0.5315,$$

while for the quadratic, the fitted values are

$$\frac{3133}{1163}, \quad \frac{18941}{5815}, \quad \frac{51361}{11630}, \quad \frac{6534}{1163}, \quad \frac{79819}{11630}, \quad \frac{47399}{5815}.$$

So the residuals are

$$\begin{aligned} 3 - \frac{3133}{1163} &= \frac{356}{1163}, & 3 - \frac{18941}{5815} &= -\frac{1496}{5815}, & 4 - \frac{51361}{11630} &= -\frac{4841}{11630}, \\ 6 - \frac{6534}{1163} &= \frac{444}{1163}, & 7 - \frac{79819}{11630} &= \frac{1591}{11630}, & 8 - \frac{47399}{5815} &= -\frac{879}{5815}. \end{aligned}$$

Therefore,

$$\begin{aligned}\text{SSE for the quadratic} &= \left(\frac{356}{1163}\right)^2 + \left(-\frac{1496}{5815}\right)^2 + \left(-\frac{4841}{11630}\right)^2 \\ &\quad + \left(\frac{444}{1163}\right)^2 + \left(\frac{1591}{11630}\right)^2 + \left(-\frac{879}{5815}\right)^2 \\ &= \frac{6053}{11630} \approx 0.5205.\end{aligned}$$

So the quadratic is technically a better least squares fit, but not by enough to strongly suggest that the true model should be quadratic. The interpolating polynomial is the one I would trust the least as a theoretical model, because it is built to hit every data point exactly, including any experimental noise. So, the ranking is

line first, quadratic second, interpolating polynomial last.

Essentially, the latter 2 models are overfitting to the training data and won't be stable for test data if this training trend is representative of the test trend.

PROBLEM 2

Let $q(t)$ denote the quadratic interpolating polynomial that goes through the data points $(t_0, y_0), (t_1, y_1), (t_2, y_2)$.

- (a) Find and simplify the coefficients a, b, c of $q(t) = at^2 - bt + c$.
- (b) What must be true for $q(t)$ to have a minimum? A maximum?
- (c) Show that the minimizing/maximizing value of $q(t)$ is at

$$t^* = \frac{m_0 s_1 - m_1 s_0}{s_1 - s_0}$$

where

$$s_0 = \frac{y_1 - y_0}{t_1 - t_0}, \quad s_1 = \frac{y_2 - y_1}{t_2 - t_1}, \quad m_0 = \frac{t_0 + t_1}{2}, \quad m_1 = \frac{t_1 + t_2}{2}.$$

- (d) Give the minimum/maximum value $q(t^*)$.

- (a) Since the interpolating polynomial is

$$q(t) = at^2 - bt + c,$$

and it passes through $(t_0, y_0), (t_1, y_1), (t_2, y_2)$, the coefficients satisfy the linear system

$$\begin{pmatrix} t_0^2 & -t_0 & 1 \\ t_1^2 & -t_1 & 1 \\ t_2^2 & -t_2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ y_2 \end{pmatrix}.$$

Equivalently, the augmented matrix is

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ t_1^2 & -t_1 & 1 & y_1 \\ t_2^2 & -t_2 & 1 & y_2 \end{array} \right).$$

Now, subtract the first row from the second, and subtract the second row from the third:

$$R_2 \leftarrow R_2 - R_1, \quad R_3 \leftarrow R_3 - R_2.$$

This gives

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ t_1^2 - t_0^2 & -(t_1 - t_0) & 0 & y_1 - y_0 \\ t_2^2 - t_1^2 & -(t_2 - t_1) & 0 & y_2 - y_1 \end{array} \right).$$

Since

$$t_1^2 - t_0^2 = (t_1 - t_0)(t_1 + t_0), \quad t_2^2 - t_1^2 = (t_2 - t_1)(t_2 + t_1),$$

we can divide the second row by $t_1 - t_0$ and the third row by $t_2 - t_1$:

$$R_2 \leftarrow \frac{1}{t_1 - t_0} R_2, \quad R_3 \leftarrow \frac{1}{t_2 - t_1} R_3.$$

Then,

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ t_1 + t_0 & -1 & 0 & \frac{y_1 - y_0}{t_1 - t_0} \\ t_2 + t_1 & -1 & 0 & \frac{y_2 - y_1}{t_2 - t_1} \end{array} \right).$$

To keep the notation shorter, I should let

$$s_0 = \frac{y_1 - y_0}{t_1 - t_0}, \quad s_1 = \frac{y_2 - y_1}{t_2 - t_1}.$$

So the augmented matrix becomes

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ t_1 + t_0 & -1 & 0 & s_0 \\ t_2 + t_1 & -1 & 0 & s_1 \end{array} \right).$$

Now, subtract the second row from the third:

$$R_3 \leftarrow R_3 - R_2.$$

Then,

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ t_1 + t_0 & -1 & 0 & s_0 \\ t_2 - t_0 & 0 & 0 & s_1 - s_0 \end{array} \right).$$

Now, divide the third row by $t_2 - t_0$:

$$R_3 \leftarrow \frac{1}{t_2 - t_0} R_3.$$

This gives

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ t_1 + t_0 & -1 & 0 & s_0 \\ 1 & 0 & 0 & \frac{s_1 - s_0}{t_2 - t_0} \end{array} \right).$$

So we can already read off

$$a = \frac{s_1 - s_0}{t_2 - t_0}.$$

Next, eliminate the a -entry from the second row:

$$R_2 \leftarrow R_2 - (t_1 + t_0)R_3.$$

Then,

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ 0 & -1 & 0 & s_0 - (t_1 + t_0)a \\ 1 & 0 & 0 & a \end{array} \right).$$

Finally, multiply the second row by -1 :

$$R_2 \leftarrow -R_2,$$

so

$$\left(\begin{array}{ccc|c} t_0^2 & -t_0 & 1 & y_0 \\ 0 & 1 & 0 & (t_1 + t_0)a - s_0 \\ 1 & 0 & 0 & a \end{array} \right).$$

Thus,

$$b = (t_1 + t_0)a - s_0.$$

If we let

$$m_0 = \frac{t_0 + t_1}{2}, \quad m_1 = \frac{t_1 + t_2}{2},$$

then this is the same as

$$b = 2am_0 - s_0 = 2am_1 - s_1.$$

To get c , eliminate the a and b entries from the first row:

$$R_1 \leftarrow R_1 - t_0^2 R_3, \quad R_1 \leftarrow R_1 + t_0 R_2.$$

This gives

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & y_0 - t_0^2 a + t_0 b \\ 0 & 1 & 0 & b \\ 1 & 0 & 0 & a \end{array} \right),$$

so

$$c = y_0 - t_0^2 a + t_0 b.$$

Therefore, the coefficients are

$$a = \frac{s_1 - s_0}{t_2 - t_0}, \quad b = 2am_0 - s_0, \quad c = y_0 - at_0^2 + bt_0.$$

(b) From part (a), we already know

$$a = \frac{s_1 - s_0}{t_2 - t_0}.$$

Since

$$q(t) = at^2 - bt + c,$$

its derivative is

$$q'(t) = 2at - b.$$

So the graph is an upward-opening parabola if $a > 0$ and a downward-opening parabola if $a < 0$. That means:

$$\text{minimum if } a > 0, \quad \text{maximum if } a < 0.$$

So, in terms of the quantities from part (a),

$$\text{minimum if } \frac{s_1 - s_0}{t_2 - t_0} > 0, \quad \text{maximum if } \frac{s_1 - s_0}{t_2 - t_0} < 0.$$

If $a = 0$, then the polynomial is not really quadratic anymore, so there is no genuine parabolic minimum or maximum to talk about.

(c) The critical point happens when

$$q'(t) = 2at - b = 0,$$

so

$$t^* = \frac{b}{2a}.$$

Now, use the coefficient relation from part (a),

$$b = 2am_0 - s_0 = 2am_1 - s_1.$$

Rearranging gives

$$s_0 = 2am_0 - b, \quad s_1 = 2am_1 - b.$$

Subtract these:

$$s_1 - s_0 = 2a(m_1 - m_0).$$

Also,

$$m_0 s_1 - m_1 s_0 = m_0(2am_1 - b) - m_1(2am_0 - b).$$

The $2am_0m_1$ terms cancel, so this becomes

$$m_0 s_1 - m_1 s_0 = b(m_1 - m_0).$$

Therefore,

$$\frac{m_0 s_1 - m_1 s_0}{s_1 - s_0} = \frac{b(m_1 - m_0)}{2a(m_1 - m_0)} = \frac{b}{2a} = t^*.$$

So indeed,

$$t^* = \frac{m_0 s_1 - m_1 s_0}{s_1 - s_0}.$$

(d) Using the same coefficients from part (a), complete the square:

$$q(t) = at^2 - bt + c = a \left(t - \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}.$$

Since

$$t^* = \frac{b}{2a},$$

we get

$$q(t^*) = c - \frac{b^2}{4a}.$$

So the minimum or maximum value is

$$\boxed{q(t^*) = c - \frac{b^2}{4a}}.$$

If $a > 0$, this is the minimum value. If $a < 0$, this is the maximum value.

PROBLEM 3

This problem sits in the context of approximating data points $(t_1, y_1), (t_2, y_2), \dots, (t_m, y_m)$ with polynomials. To find the coefficients of the least squares approximating polynomial, we developed an orthogonal basis $q_0(t), q_1(t), q_2(t), \dots$ and used that to get an orthogonal set of sampling points of the data $\mathbf{q}_0, \mathbf{q}_1, \mathbf{q}_2, \dots$:

$$\begin{aligned} q_0(t) &= 1, & \mathbf{q}_0 &= \mathbf{t}_0, & \|\mathbf{q}_0\|^2 &= 1, \\ q_1(t) &= t, & \mathbf{q}_1 &= \mathbf{t}_1, & \|\mathbf{q}_1\|^2 &= \bar{t}^2, \\ q_2(t) &= t^2 - \bar{t}^2, & \mathbf{q}_2 &= \mathbf{t}_2 - \bar{t}^2 \mathbf{t}_0, & \|\mathbf{q}_2\|^2 &= \bar{t}^4 - \left(\bar{t}^2\right)^2, \\ q_3(t) &= t^3 - \frac{\bar{t}^4}{\bar{t}^2} t, & \mathbf{q}_3 &= \mathbf{t}_3 - \frac{\bar{t}^4}{\bar{t}^2} \mathbf{t}_1, & \|\mathbf{q}_3\|^2 &= \bar{t}^6 - \frac{\left(\bar{t}^4\right)^2}{\bar{t}^2}, \end{aligned}$$

where

$$\mathbf{t}_k = (t_1^k, t_2^k, \dots, t_m^k)^T \quad \text{and} \quad \bar{t}^k = \frac{1}{m} \sum_{i=1}^m t_i^k.$$

When we generated these polynomials, we made some simplifying assumptions, namely that the t -coordinates of the data points are uniformly distributed about $t = 0$. As long as the data is uniformly distributed, we can convert the least squares problem to the symmetric form as follows.

- (a) Assume $t_{i+1} - t_i = h$ for all $i = 1, \dots, m-1$. Show that the translated sample points $s_i = t_i - \bar{t}$, where

$$\bar{t} = \frac{1}{m} \sum_{i=1}^m t_i,$$

are symmetric around 0.

- (b) Suppose that $q(s)$ is the least squares polynomial for the data points (s_i, y_i) . Show that $p(t) = q(1 - t)$ is the least squares polynomial for the original data (t_i, y_i) .
- (c) Apply this method to find the least squares polynomials of degrees 1, 2, and 3 for the following data:

$$\begin{array}{c|ccccc} t_i & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline y_i & -8 & -6 & -4 & -1 & 1 & 3 \end{array}.$$

- (a) Since the t_i are equally spaced, each new point is obtained by adding the same step size h . So

$$t_i = t_1 + (i-1)h.$$

The last point is

$$t_m = t_1 + (m-1)h,$$

so the midpoint of the first and last points is

$$\frac{t_1 + t_m}{2} = \frac{t_1 + t_1 + (m-1)h}{2} = t_1 + \frac{m-1}{2}h.$$

For equally spaced points, this midpoint is also the average value, so

$$\bar{t} = t_1 + \frac{m-1}{2}h.$$

Therefore,

$$s_i = t_i - \bar{t} = t_1 + (i-1)h - \left(t_1 + \frac{m-1}{2}h\right) = \left(i - \frac{m+1}{2}\right)h.$$

Now, check the point paired with s_i , namely s_{m+1-i} :

$$s_{m+1-i} = \left((m+1-i) - \frac{m+1}{2} \right) h = \left(\frac{m+1}{2} - i \right) h = -s_i.$$

So the translated points come in opposite pairs, which is exactly what it means to be symmetric around 0.

(b) From part (a), the shifted variable is

$$s = t - \bar{t},$$

so the corresponding polynomial for the original variable should be

$$p(t) = q(t - \bar{t}).$$

The shift coming from part (a) depends on $\bar{t}$, not on the fixed number 1, it seems. Also, to recover the original data, we need

$$p(t_i) = q(s_i) = q(t_i - \bar{t})$$

at every sample point. But $q(1 - t_i)$ would usually give different inputs, so it would not match the shifted least squares problem unless we happened to be in a very special case. So I'm going to assume that it was supposed to be

$$p(t) = q(t - \bar{t}).$$

Now, let $p(t)$ be any polynomial of the same degree, and define

$$r(s) = p(s + \bar{t}).$$

Then, at the sample points, since $s_i = t_i - \bar{t}$, we have

$$r(s_i) = p(t_i).$$

So the least squares error for the original data is

$$\sum_{i=1}^m (y_i - p(t_i))^2 = \sum_{i=1}^m (y_i - r(s_i))^2.$$

But $q(s)$ is assumed to be the least squares polynomial for the shifted data (s_i, y_i) , so

$$\sum_{i=1}^m (y_i - q(s_i))^2 \leq \sum_{i=1}^m (y_i - r(s_i))^2.$$

Replace $r(s_i)$ by $p(t_i)$:

$$\sum_{i=1}^m (y_i - q(s_i))^2 \leq \sum_{i=1}^m (y_i - p(t_i))^2.$$

Now, define

$$p(t) = q(t - \bar{t}).$$

Then

$$p(t_i) = q(s_i),$$

so $p(t)$ is the least squares polynomial for the original data.

(c) For this data,

$$t_1 = 1, \quad t_2 = 2, \quad t_3 = 3, \quad t_4 = 4, \quad t_5 = 5, \quad t_6 = 6.$$

Their average is

$$\bar{t} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{7}{2}.$$

So the translated points are

$$s_i = t_i - \frac{7}{2},$$

namely

$$-\frac{5}{2}, \quad -\frac{3}{2}, \quad -\frac{1}{2}, \quad \frac{1}{2}, \quad \frac{3}{2}, \quad \frac{5}{2}.$$

These are symmetric around 0, just as we derived in part (a), so it makes sense to shift; once the sample points are centered around 0, we can use the orthogonal basis formulas from the problem statement directly. Then, after finding the least squares polynomial in the shifted variable s , part (b) tells us how to return to the original variable t , namely by substituting

$$s = t - \frac{7}{2}.$$

Let

$$\mathbf{s}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{s}_1 = \begin{pmatrix} -\frac{5}{2} \\ -\frac{3}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ \frac{3}{2} \\ \frac{5}{2} \end{pmatrix}.$$

The next sample vectors are

$$\mathbf{s}_2 = \begin{pmatrix} \frac{25}{4} \\ \frac{9}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{9}{4} \\ \frac{25}{4} \end{pmatrix}, \quad \mathbf{s}_3 = \begin{pmatrix} -\frac{125}{8} \\ -\frac{27}{8} \\ -\frac{1}{8} \\ \frac{1}{8} \\ \frac{27}{8} \\ \frac{125}{8} \end{pmatrix}.$$

To build the orthogonal vectors from the formulas in the problem statement, first compute the needed sample averages:

$$\overline{s^2} = \frac{35}{12}, \quad \overline{s^4} = \frac{707}{48}, \quad \overline{s^6} = \frac{16355}{192}.$$

So the orthogonal basis polynomials become

$$q_0(s) = 1, \quad q_1(s) = s, \quad q_2(s) = s^2 - \frac{35}{12},$$

$$q_3(s) = s^3 - \frac{\overline{s^4}}{\overline{s^2}} s = s^3 - \frac{101}{20} s.$$

The corresponding orthogonal sample vectors are

$$\mathbf{q}_0 = \mathbf{s}_0, \quad \mathbf{q}_1 = \mathbf{s}_1, \quad \mathbf{q}_2 = \mathbf{s}_2 - \frac{35}{12} \mathbf{s}_0 = \begin{pmatrix} \frac{10}{3} \\ -\frac{2}{3} \\ -\frac{8}{3} \\ -\frac{8}{3} \\ -\frac{2}{3} \\ \frac{10}{3} \end{pmatrix},$$

$$\mathbf{q}_3 = \mathbf{s}_3 - \frac{101}{20}\mathbf{s}_1 = \begin{pmatrix} -3 \\ \frac{21}{5} \\ \frac{12}{5} \\ -\frac{12}{5} \\ -\frac{21}{5} \\ 3 \end{pmatrix}.$$

$$\vec{y} = \begin{pmatrix} -8 \\ -6 \\ -4 \\ -1 \\ 1 \\ 3 \end{pmatrix}.$$

Now, use the orthogonal-basis formula:

$$p_n(s) = \sum_{k=0}^n \frac{\langle \vec{y}, \mathbf{q}_k \rangle}{\|\mathbf{q}_k\|^2} q_k(s).$$

So we first compute the needed norms:

$$\|\mathbf{q}_0\|^2 = 1,$$

$$\|\mathbf{q}_1\|^2 = \frac{1}{6} \left[\left(\frac{5}{2}\right)^2 + \left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2 + \left(\frac{5}{2}\right)^2 \right] = \frac{35}{12},$$

$$\|\mathbf{q}_2\|^2 = \frac{1}{6} \left[\left(\frac{10}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(-\frac{8}{3}\right)^2 + \left(-\frac{8}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{10}{3}\right)^2 \right] = \frac{56}{9},$$

$$\|\mathbf{q}_3\|^2 = \frac{1}{6} \left[9 + \left(\frac{21}{5}\right)^2 + \left(\frac{12}{5}\right)^2 + \left(-\frac{12}{5}\right)^2 + \left(-\frac{21}{5}\right)^2 + 9 \right] = \frac{54}{5}.$$

The corresponding inner products are

$$\langle \vec{y}, \mathbf{q}_0 \rangle = -\frac{5}{2}, \quad \langle \vec{y}, \mathbf{q}_1 \rangle = \frac{79}{12}, \quad \langle \vec{y}, \mathbf{q}_2 \rangle = 0, \quad \langle \vec{y}, \mathbf{q}_3 \rangle = -\frac{3}{5}.$$

For degree 1, we keep the q_0 and q_1 terms:

$$p_1^*(s) = \frac{\langle \vec{y}, \mathbf{q}_0 \rangle}{\|\mathbf{q}_0\|^2} q_0(s) + \frac{\langle \vec{y}, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2} q_1(s) = -\frac{5}{2} + \frac{79}{35}s.$$

Now, substitute $s = t - \frac{7}{2}$:

$$p_1(t) = -\frac{5}{2} + \frac{79}{35} \left(t - \frac{7}{2} \right) = \frac{79}{35}t - \frac{52}{5}.$$

For degree 2, we add the q_2 term:

$$p_2^*(s) = p_1^*(s) + \frac{\langle \vec{y}, \mathbf{q}_2 \rangle}{\|\mathbf{q}_2\|^2} q_2(s) = -\frac{5}{2} + \frac{79}{35}s + \frac{0}{56/9} \left(s^2 - \frac{35}{12} \right).$$

So

$$p_2^*(s) = -\frac{5}{2} + \frac{79}{35}s,$$

and therefore

$$p_2(t) = \frac{79}{35}t - \frac{52}{5}.$$

For degree 3, we add the q_3 term:

$$p_3^*(s) = p_2^*(s) + \frac{\langle \vec{y}, \mathbf{q}_3 \rangle}{\|\mathbf{q}_3\|^2} q_3(s) = -\frac{5}{2} + \frac{79}{35}s + \frac{-3/5}{54/5} \left(s^3 - \frac{101}{20}s \right).$$

Since

$$\frac{-3/5}{54/5} = -\frac{1}{18},$$

this becomes

$$p_3^*(s) = -\frac{5}{2} + \frac{79}{35}s - \frac{1}{18} \left(s^3 - \frac{101}{20}s \right).$$

After simplifying,

$$p_3^*(s) = -\frac{5}{2} + \frac{1279}{504}s - \frac{1}{18}s^3.$$

Now, substitute $s = t - \frac{7}{2}$:

$$p_3(t) = -\frac{1}{18}t^3 + \frac{7}{12}t^2 + \frac{125}{252}t - 9.$$

So the least squares polynomials are

$$p_1(t) = \frac{79}{35}t - \frac{52}{5},$$

$$p_2(t) = \frac{79}{35}t - \frac{52}{5},$$

$$p_3(t) = -\frac{1}{18}t^3 + \frac{7}{12}t^2 + \frac{125}{252}t - 9.$$

PROBLEM 4

Use the spectral factorization to diagonalize the following quadratic forms.

(a) $x^2 + 8xz + y^2 + 6yz + z^2$

(b) $6x^2 - 8xy + 2xz + 6y^2 - 2yz + 11z^2$

(a) The quadratic form is

$$Q(x, y, z) = x^2 + 8xz + y^2 + 6yz + z^2,$$

so the associated symmetric matrix (from our "trick"; I still don't know the official name of this) is

$$A = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 4 & 3 & 1 \end{pmatrix}.$$

Since A is symmetric, the spectral theorem says its eigenvalues are real and its eigenvectors for different eigenvalues are orthogonal. Using the spectral factorization, we find the eigenvalues

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & 0 & 4 \\ 0 & 1 - \lambda & 3 \\ 4 & 3 & 1 - \lambda \end{vmatrix} = -(\lambda - 6)(\lambda - 1)(\lambda + 4).$$

So the eigenvalues are

$$6, \quad 1, \quad -4$$

Now, solve $(A - \lambda I)\vec{v} = 0$ for each one. For $\lambda = 6$,

$$A - 6I = \begin{pmatrix} -5 & 0 & 4 \\ 0 & -5 & 3 \\ 4 & 3 & -5 \end{pmatrix},$$

so the equations reduce to

$$-5x + 4z = 0, \quad -5y + 3z = 0.$$

Taking $z = 5$, we get $x = 4$ and $y = 3$, so one eigenvector is

$$\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}.$$

For $\lambda = 1$,

$$A - I = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 0 & 3 \\ 4 & 3 & 0 \end{pmatrix},$$

so

$$4z = 0, \quad 3z = 0, \quad 4x + 3y = 0.$$

Thus, $z = 0$, and taking $y = 4$ gives $x = -3$. So one eigenvector is

$$\begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix}.$$

For $\lambda = -4$,

$$A + 4I = \begin{pmatrix} 5 & 0 & 4 \\ 0 & 5 & 3 \\ 4 & 3 & 5 \end{pmatrix},$$

so the equations reduce to

$$5x + 4z = 0, \quad 5y + 3z = 0.$$

Taking $z = 5$, we get $x = -4$ and $y = -3$, so one eigenvector is

$$\begin{pmatrix} -4 \\ -3 \\ 5 \end{pmatrix}.$$

So the corresponding eigenvectors are

$$\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}, \quad \begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -4 \\ -3 \\ 5 \end{pmatrix}.$$

After normalizing, we get the orthonormal eigenvectors

$$\vec{u}_1 = \frac{1}{5\sqrt{2}} \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}, \quad \vec{u}_2 = \frac{1}{5} \begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix}, \quad \vec{u}_3 = \frac{1}{5\sqrt{2}} \begin{pmatrix} -4 \\ -3 \\ 5 \end{pmatrix}.$$

These satisfy

$$A\vec{u}_1 = 6\vec{u}_1, \quad A\vec{u}_2 = \vec{u}_2, \quad A\vec{u}_3 = -4\vec{u}_3.$$

Now, place these orthonormal eigenvectors into the columns of a matrix:

$$P = \begin{pmatrix} \frac{4}{5\sqrt{2}} & -\frac{3}{5} & -\frac{4}{5\sqrt{2}} \\ \frac{3}{5\sqrt{2}} & \frac{4}{5} & -\frac{3}{5\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Because the columns are orthonormal, P is an orthogonal matrix, so

$$P^T P = I.$$

Next, place the eigenvalues on the diagonal:

$$\Sigma = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -4 \end{pmatrix}.$$

If we multiply A by P , column by column we get

$$AP = (A\vec{u}_1 \quad A\vec{u}_2 \quad A\vec{u}_3) = (6\vec{u}_1 \quad \vec{u}_2 \quad -4\vec{u}_3).$$

But that is exactly the same as multiplying P by D :

$$P\Sigma = (6\vec{u}_1 \quad \vec{u}_2 \quad -4\vec{u}_3).$$

So

$$AP = P\Sigma.$$

Now, multiply on the left by P^T :

$$P^T AP = \Sigma.$$

So the diagonalization in matrix form is

$$\boxed{A = P\Sigma P^T}.$$

If we also write the quadratic form in the new coordinates

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = P^T \begin{pmatrix} x \\ y \\ z \end{pmatrix},$$

that is,

$$u = \frac{4x + 3y + 5z}{5\sqrt{2}}, \quad v = \frac{-3x + 4y}{5}, \quad w = \frac{-4x - 3y + 5z}{5\sqrt{2}},$$

then

$$Q = 6u^2 + v^2 - 4w^2.$$

(b) Now, take

$$Q(x, y, z) = 6x^2 - 8xy + 2xz + 6y^2 - 2yz + 11z^2.$$

The associated symmetric matrix is

$$A = \begin{pmatrix} 6 & -4 & 1 \\ -4 & 6 & -1 \\ 1 & -1 & 11 \end{pmatrix}.$$

Since this matrix is also symmetric, the spectral theorem applies again. This time,

$$\det(A - \lambda I) = -(\lambda - 2)(\lambda - 9)(\lambda - 12),$$

so the eigenvalues are

$$2, \quad 9, \quad 12.$$

Now, solve $(A - \lambda I)\vec{v} = 0$. For $\lambda = 2$,

$$A - 2I = \begin{pmatrix} 4 & -4 & 1 \\ -4 & 4 & -1 \\ 1 & -1 & 9 \end{pmatrix},$$

which reduces to

$$x - y = 0, \quad z = 0.$$

So one eigenvector is

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},$$

For $\lambda = 9$,

$$A - 9I = \begin{pmatrix} -3 & -4 & 1 \\ -4 & -3 & -1 \\ 1 & -1 & 2 \end{pmatrix},$$

which reduces to

$$x + z = 0, \quad y - z = 0.$$

Taking $z = 1$ gives one eigenvector

$$\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix},$$

For $\lambda = 12$,

$$A - 12I = \begin{pmatrix} -6 & -4 & 1 \\ -4 & -6 & -1 \\ 1 & -1 & -1 \end{pmatrix},$$

which reduces to

$$x - 2z = 0, \quad y + 2z = 0.$$

Taking $z = \frac{1}{2}$ gives $x = 1$ and $y = -1$, so one eigenvector is

$$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

So the corresponding eigenvectors are

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Normalize them:

$$\vec{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{u}_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad \vec{u}_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Let

$$P = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{pmatrix}.$$

Let

$$\Sigma = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 12 \end{pmatrix}.$$

Then, again, P is orthogonal, and

$$P^T A P = \Sigma.$$

So the diagonalization in matrix form is

$$\boxed{A = P \Sigma P^T}.$$

If we also write the new coordinates as

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = P^T \begin{pmatrix} x \\ y \\ z \end{pmatrix},$$

meaning

$$u = \frac{x+y}{\sqrt{2}}, \quad v = \frac{-x+y+z}{\sqrt{3}}, \quad w = \frac{x-y+2z}{\sqrt{6}},$$

then the quadratic form becomes

$$Q = 2u^2 + 9v^2 + 12w^2.$$

PROBLEM 5

Find the singular values and use them to find the condition number of the following matrices. Which would you classify as ill-conditioned? Remember, the bigger the condition number the closer to singular the matrix is.

(a) $\begin{pmatrix} -0.999 & 0.341 \\ -1.001 & 0.388 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 2 \\ 1.001 & 1.9997 \end{pmatrix}$

(c) $\begin{pmatrix} -1 & 3 & 4 \\ 2 & 10 & 6 \\ 1 & 2 & -3 \end{pmatrix}$

(d) $\begin{pmatrix} 72 & 96 & 103 \\ 42 & 55 & 59 \\ 67 & 95 & 102 \end{pmatrix}$

- (a) For each matrix A , the singular values are the square roots of the eigenvalues of $A^T A$ (I don't have a fancy TI-Nspire). My TI-84 calculator gives the numerical values below. For

$$A = \begin{pmatrix} -0.999 & 0.341 \\ -1.001 & 0.388 \end{pmatrix},$$

the singular values are approximately

$$\sigma_1 \approx 1.50528, \quad \sigma_2 \approx 0.030739.$$

So the condition number is

$$\kappa(A) = \frac{\sigma_1}{\sigma_2} \approx 48.97.$$

- (b) For

$$A = \begin{pmatrix} 1 & 2 \\ 1.001 & 1.9997 \end{pmatrix},$$

the singular values are approximately

$$\sigma_1 \approx 3.16240, \quad \sigma_2 \approx 0.000727295.$$

So

$$\kappa(A) \approx \frac{3.16240}{0.000727295} \approx 4348.17.$$

- (c) For

$$A = \begin{pmatrix} -1 & 3 & 4 \\ 2 & 10 & 6 \\ 1 & 2 & -3 \end{pmatrix},$$

the singular values are approximately

$$\sigma_1 \approx 12.65565, \quad \sigma_2 \approx 4.34391, \quad \sigma_3 \approx 0.982263.$$

Therefore,

$$\kappa(A) \approx \frac{12.65565}{0.982263} \approx 12.88.$$

(d) For

$$A = \begin{pmatrix} 72 & 96 & 103 \\ 42 & 55 & 59 \\ 67 & 95 & 102 \end{pmatrix},$$

the singular values are approximately

$$\sigma_1 \approx 239.13786, \quad \sigma_2 \approx 3.17545, \quad \sigma_3 \approx 0.00131688.$$

So the condition number is

$$\kappa(A) \approx \frac{239.13786}{0.00131688} \approx 181593.98.$$

Overall,

(b) and (d)

seem ill-conditioned. Their condition numbers are huge, especially (d). Matrix (a) is a lot more sensitive than a well-conditioned matrix, but it is still nowhere near as extreme as (b) and (d). Matrix (c) looks reasonably well-conditioned compared with the others.

PROBLEM 6

Use the pseudoinverse to find the least squares solution of minimal norm to the following linear systems.

(a)

$$\begin{aligned} x + y &= 1, \\ 3x + 3y &= -2 \end{aligned}$$

(b)

$$\begin{aligned} x + y + z &= 5, \\ 2x - y + z &= 2 \end{aligned}$$

(c)

$$\begin{aligned} x - 3y &= 2, \\ 2x + y &= -1, \\ x + y &= 0 \end{aligned}$$

(a) The system is

$$\begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Call the coefficient matrix A . Since both columns of A are the same, it is convenient to write

$$A = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}.$$

Let

$$\vec{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Their lengths are

$$\|\vec{a}\| = \sqrt{1^2 + 3^2} = \sqrt{10}, \quad \|\vec{b}\| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

So the corresponding unit vectors are

$$\vec{u} = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \vec{v} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Equivalently,

$$\vec{a} = \sqrt{10} \vec{u}, \quad \vec{b} = \sqrt{2} \vec{v}.$$

Then,

$$A = \vec{a} \vec{b}^T = (\sqrt{10} \vec{u})(\sqrt{2} \vec{v})^T = \sqrt{10} \sqrt{2} \vec{u} \vec{v}^T = (2\sqrt{5}) \vec{u} \vec{v}^T.$$

So this is an SVD of the form

$$A = U \Sigma V^T$$

with singular value

$$\sigma_1 = 2\sqrt{5}.$$

More explicitly,

$$U = \begin{pmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{pmatrix}, \quad \Sigma = (2\sqrt{5}), \quad V^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Therefore,

$$A^+ = V \Sigma^+ U^T \quad \Sigma^+ = \left(\frac{1}{2\sqrt{5}} \right).$$

So

$$A^+ = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{2\sqrt{5}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{pmatrix}.$$

First, multiply the first two factors:

$$V\Sigma^+ = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{2\sqrt{5}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2\sqrt{10}} \\ \frac{1}{2\sqrt{10}} \end{pmatrix}.$$

Then,

$$A^+ = (V\Sigma^+)U^T = \begin{pmatrix} \frac{1}{2\sqrt{10}} \\ \frac{1}{2\sqrt{10}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{pmatrix} = \frac{1}{20} \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{20} & \frac{3}{20} \\ \frac{1}{20} & \frac{3}{20} \end{pmatrix}.$$

Therefore, the least squares solution of minimal norm is

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^+ \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} \frac{1}{20} & \frac{3}{20} \\ \frac{1}{20} & \frac{3}{20} \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} \frac{1}{20} - \frac{6}{20} \\ \frac{1}{20} - \frac{6}{20} \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{pmatrix}.$$

So the minimal norm least squares solution is

$$\boxed{\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{pmatrix}}.$$

(b) Now,

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}.$$

This matrix has rank 2, which is the number of rows, so

$$A^+ = A^T(AA^T)^{-1}.$$

You might be wondering where this came from; I read some of the chapters surrounding SVD and came across special-case formulas for full "column rank" and "row rank" cases. Here, the matrix A is 2×3 , so AA^T is a 2×2 matrix while $A^T A$ is a 3×3 matrix. Since the rank is 2, the two rows of A are independent. That means the 2×2 matrix AA^T is regular, so it makes sense to invert AA^T . On the other hand, $A^T A$ is 3×3 but the rank is only 2, so $A^T A$ cannot be invertible. Now, first compute

$$AA^T = \begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}.$$

Its determinant is

$$\det(AA^T) = 3 \cdot 6 - 2 \cdot 2 = 14,$$

so its inverse is

$$(AA^T)^{-1} = \frac{1}{14} \begin{pmatrix} 6 & -2 \\ -2 & 3 \end{pmatrix}.$$

So

$$A^+ = \begin{pmatrix} 1 & 2 \\ 1 & -1 \\ 1 & 1 \end{pmatrix} \frac{1}{14} \begin{pmatrix} 6 & -2 \\ -2 & 3 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 1 & 2 \\ 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 6 & -2 \\ -2 & 3 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 2 & 4 \\ 8 & -5 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{4}{7} & -\frac{5}{14} \\ \frac{2}{7} & \frac{1}{14} \end{pmatrix}.$$

Now, multiply by $\vec{b}$:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^+ \vec{b} = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{4}{7} & -\frac{5}{14} \\ \frac{2}{7} & \frac{1}{14} \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 2 & 4 \\ 8 & -5 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 18 \\ 30 \\ 22 \end{pmatrix} = \begin{pmatrix} \frac{5}{7} + \frac{4}{7} \\ \frac{20}{7} - \frac{10}{14} \\ \frac{10}{7} + \frac{2}{14} \end{pmatrix} = \begin{pmatrix} \frac{9}{7} \\ \frac{15}{7} \\ \frac{11}{7} \end{pmatrix}.$$

So the least squares solution of minimal norm is

$$\boxed{\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{9}{7} \\ \frac{15}{7} \\ \frac{11}{7} \end{pmatrix}}.$$

In fact, this system is consistent, so this is an exact solution too.

(c) Here,

$$A = \begin{pmatrix} 1 & -3 \\ 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}.$$

This matrix has rank 2, which is the number of columns, so

$$A^+ = (A^T A)^{-1} A^T.$$

Here, the situation is the other way around. The matrix A is 3×2 , so $A^T A$ is a 2×2 matrix while AA^T is a 3×3 matrix. Since the rank is 2, the two columns of A are independent. That means the 2×2 matrix $A^T A$ is regular, so it can be inverted. But AA^T is 3×3 while the rank is still only 2, so AA^T cannot be invertible. Now,

$$A^T A = \begin{pmatrix} 1 & 2 & 1 \\ -3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 11 \end{pmatrix}.$$

Since the columns are orthogonal, the computation becomes much nicer than usual. Then

$$(A^T A)^{-1} = \begin{pmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{11} \end{pmatrix},$$

and so

$$A^+ = \begin{pmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{11} \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ -3 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{6} \\ -\frac{3}{11} & \frac{1}{11} & \frac{1}{11} \end{pmatrix}.$$

Now, multiply by $\vec{b}$:

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^+ \vec{b} = \begin{pmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{6} \\ -\frac{3}{11} & \frac{1}{11} & \frac{1}{11} \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2}{6} - \frac{1}{3} + 0 \\ -\frac{6}{11} - \frac{1}{11} + 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} - \frac{1}{3} \\ -\frac{7}{11} \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{7}{11} \end{pmatrix}.$$

So the minimal norm least squares solution is

$$\boxed{\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{7}{11} \end{pmatrix}}.$$